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Amortized Chu-former

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Amortized Chu-former

Fast semantic handshakes

Gary Nan Tie, Nov 14, 2025

Abstract

Chu-former semantic handshakes between questions and answers are achieved by learning a pair of fibration liftings. We introduce an amortized version to reduce the computational burden at the expense of approximate explanation.

Amortized Chu-Former

11/14/25

Chu-Former recap: Latent space $\mathbb{R}^d$,queries $q_i \in Q \subseteq \mathbb{R}^d$, $i \in [n]$,key-value pairs $(k_j, v_j) \in K \times V \subseteq \mathbb{R}^d \times \mathbb{R}^d$, $j \in [m]$,and score a similarity function between keys and queries,
like a reproducing kernel.For $j \in [m]$, let $\alpha_{ij} \triangleq \underset{i \in [n]}{\text{softmax}} \text{score}(k_j, q_i)$.Learn NN $p: K \rightarrow V$ such that $p(k_j) =_\epsilon v_j$, $j \in [m]$ and learn NN $d: K \rightarrow Q$ such that

$$d(k_j) =_\epsilon \sum_{i=1}^n \alpha_{ij} q_i, \quad j \in [m].$$

For $i \in [n]$, let $\beta_{ij} \triangleq \underset{j \in [m]}{\text{softmax}} \text{score}(q_i, k_j)$ and $j^* \triangleq \arg \max_j \beta_{ij}$.

Given $d: K \rightarrow Q$ and $p: K \rightarrow V$,

learn NN $\tau: K \otimes Q \rightarrow V \otimes K$ such that

$$(\dagger) \quad \begin{array}{ccc} K \otimes K & \xrightarrow{p \otimes 1} & V \otimes K \\ 1 \otimes d \downarrow & \tau \nearrow & \downarrow 1 \otimes d \\ K \otimes Q & \xrightarrow{p \otimes 1} & V \otimes Q \end{array} \quad \text{commutes up to } =_{\varepsilon}.$$

Learn $c: Q \rightarrow K$ such that $c(q_i) =_{\varepsilon} \pi_2 \tau \begin{bmatrix} k_{j^*} \\ q_i \end{bmatrix}$

then $d(c(q_i)) =_{\varepsilon} q_i$, $i \in [n]$.

Similarly, learn NN $\sigma: K \otimes V \rightarrow Q \otimes K$ such that

$$(\ddagger) \quad \begin{array}{ccc} K \otimes K & \xrightarrow{d \otimes 1} & Q \otimes K \\ 1 \otimes p \downarrow & \sigma \nearrow & \downarrow 1 \otimes p \\ K \otimes V & \xrightarrow{d \otimes 1} & Q \otimes V \end{array} \quad \text{commutes up to } =_{\varepsilon}.$$

Learn $s: V \rightarrow K$ such that $s(v_j) =_{\varepsilon} \pi_2 \sigma \begin{bmatrix} k_j \\ v_j \end{bmatrix}$

then $p(s(v_j)) =_{\varepsilon} v_j$, $j \in [m]$.

Let $m: K \times K \rightarrow \mathbb{R}^+$ be a metric on keys
in the latent space.

Define $s(V_j)$ to be the key closest to $c(q_i)$
under the metric m .

In reply to query q_i , answer value V_j .

Query updating:

For $0 < \alpha < \infty$, $\alpha \neq 1$,

$$H_\alpha(\{p_i\}_{i=1}^n) \triangleq \frac{1}{1-\alpha} \ln \left(\sum_{i=1}^n p_i^\alpha \right),$$

where $0 \leq p_i \leq 1$ and $\sum_{i=1}^n p_i = 1$.

For $\alpha = 0, 1, \infty$ define $H_\alpha \triangleq \lim_{\gamma \rightarrow \alpha} H_\gamma$,

Renyi α -entropy.

Choose hyperparameter $\alpha \geq 0$.

Define $E(\{q_i\}_{i=1}^n) \triangleq \frac{1}{n} \sum_{i=1}^n H_\alpha^i$,

where $H_\alpha^i = H_\alpha(\{\beta_{ij}\}_{j=1}^m)$

and $\beta_{ij} = \underset{j \in [m]}{\text{softmax score}}(q_i, k_j)$.

Let query update $q'_i \triangleq \sum_{j=1}^m \beta_{ij} v_j$ and

τ' be the corresponding fibration lifting

using $d'(k_j) = \varepsilon \sum_{i=1}^n \alpha'_{ij} q'_i$, $\alpha'_{ij} = \underset{i \in [n]}{\text{softmax score}}(k_j, q'_i)$,

likewise corresponding lifting $\sigma'(d')$.

For $i \in [n]$, define $\beta'_{ij} = \underset{j \in [m]}{\text{softmax score}}(q'_i, k_j)$.

Note that $E(\{q_i\}_{i=1}^n)$ depends on $\{\beta_{ij}\}$

and $E(\{q'_i\}_{i=1}^n)$ depends on $\{\beta'_{ij}\}$.

Updating process:

$$\text{If } E(\{q'_i\}_{i=1}^n) < E(\{q_i\}_{i=1}^n)$$

assign $q_i := q'_i$, $\tau := \tau'$, and $\sigma := \sigma'$ and repeat,

$$\text{until } E(\{q'_i\}_{i=1}^n) \geq E(\{q_i\}_{i=1}^n)$$

or $E(\{q'_i\}_{i=1}^n)$ is sufficiently small,

at which point return value v_j ,

in answer to updated query q'_i .

—

So query updating involves learning d' , τ' , and σ' .

To reduce the computational burden we introduce amortization of the semantic handshake.

For $\varepsilon > 0$, define $x \approx_\varepsilon y$ iff $L(x, y) \leq \varepsilon$,

where $L: \mathbb{R}^d \times \mathbb{R}^d \rightarrow \mathbb{R}^+$ is a loss function.

$$\begin{array}{ccc} K \otimes K & \xrightarrow{p \otimes 1} & V \otimes K \\ \downarrow d' & \tau' \nearrow & \downarrow 1 \otimes d' \\ K \otimes Q & \xrightarrow{p \otimes 1} & V \otimes Q \end{array} \Rightarrow d'(c'(q'_i)) \approx_\varepsilon q'_i, i \in [n].$$

So given $d': K \rightarrow Q$ and $p: K \rightarrow V$ (semantics of Q and V),

fibration lifting $\tau': K \otimes Q \rightarrow V \otimes K$ yields

a section $c': Q \rightarrow K$ of $d': K \rightarrow Q$,

in that $L(d'(c'(q'_i)), q'_i) \leq \varepsilon, \forall i \in [n]$.

Define $k_{\tilde{j}}(q'_i) \triangleq \arg \min_{j \in [m]} L(d'(k_j), q'_i)$,

an amortized proxy for $c'(q'_i)$.

Let $\tilde{\varepsilon}_i = L(d'(k_{\tilde{j}}), q'_i)$; if $\tilde{\varepsilon}_i > \varepsilon$,

then warn a user that under semantics $d': K \rightarrow Q$,

$k_{\tilde{j}}$ does not fully explain q'_i . Return value $v_{\tilde{j}} \approx_\varepsilon p(k_{\tilde{j}})$ in answer to updated query q'_i .

Recall, σ' defines a section s' of p :

$$\begin{array}{ccc}
 & d' \otimes 1 & \\
 K \otimes K & \xrightarrow{\quad} & Q \otimes K \\
 1 \otimes p \downarrow \sigma' \nearrow \downarrow 1 \otimes p & & \\
 K \otimes V & \xrightarrow{\quad} & Q \otimes V \\
 & d' \otimes 1 &
 \end{array}
 \Rightarrow p(s'(v_j)) =_\varepsilon v_j, j \in [m]$$

Semantic handshake

$$\begin{array}{ccccc}
 & & c'(q'_i) \approx_m s'(v_j) & & \\
 d' \swarrow & & & & \searrow p \\
 q'_i & & & & v_j
 \end{array}$$

versus amortized

$$\begin{array}{ccccc}
 & & k_j & & \\
 d' \swarrow & & & & \searrow p \\
 q'_i =_{\varepsilon_i} d'(k_j) & & & & v_j
 \end{array}$$

Summary For amortized query updating.

if $E(\{q'_i\}_{i=1}^n) < E(\{q_i\}_{i=1}^n)$,

compute amortization $K_{\tilde{E}}(q'_i)$, $i \in [n]$,

a proxy for $c'(q'_i)$, and return value $V_{\tilde{E}}$

in answer to updated query q'_i .

If $\tilde{E}_i > \varepsilon$, warn a user that under semantics

$d': K \rightarrow Q$, $K_{\tilde{E}}$ does not fully explain q'_i .

We have traded off the precision of learning NN

d' , τ' and σ' for the speed of amortization

and an $\tilde{E}$ -approximate semantic handshake.

□

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Gary Nan Tie, Sep 25, 2025

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Transformer attention uses statistical correlation to update queries, and fibration abduction attention on top, takes into account key-value semantics, finding a key to explain the value returned. The latter has a quantum interpretation, indeed a fibration lifting is an entanglement. Entanglements (inner loop) can be refined using a Renyi α -entropy criterion, as part of a simultaneous query and key update (outer loop). Thus the value returned then reflects refinement of both query-key similarity and key-value semantics.

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